# Concepts › Overview › Components

Kubernetes Components | Kubernetes
Kubernetes ComponentsAn overview of the key components that make up a Kubernetes cluster.This page provides a high-level overview of the essential components that make up a Kubernetes cluster.The components of a Kubernetes clusterCore ComponentsA Kubernetes cluster consists of a control plane and one or more worker nodes.
Here's a brief overview of the main components:Control Plane ComponentsManage the overall state of the cluster:kube-apiserverThe core component server that exposes the Kubernetes HTTP API.etcdConsistent and highly-available key value store for all API server data.kube-schedulerLooks for Pods not yet bound to a node, and assigns each Pod to a suitable node.kube-controller-managerRuns controllers to implement Kubernetes API behavior.cloud-controller-manager (optional)Integrates with underlying cloud provider(s).Node ComponentsRun on every node, maintaining running pods and providing the Kubernetes runtime environment:kubeletEnsures that Pods are running, including their containers.kube-proxy (optional)Maintains network rules on nodes to implement Services.Container runtimeSoftware responsible for running containers. Read
Container Runtimes to learn more.■ This item links to a third party project or product that is not part of Kubernetes itself. More informationYour cluster may require additional software on each node; for example, you might also
run systemd on a Linux node to supervise local components.AddonsAddons extend the functionality of Kubernetes. A few important examples include:DNSFor cluster-wide DNS resolution.Web UI (Dashboard)For cluster management via a web interface.Container Resource MonitoringFor collecting and storing container metrics.Cluster-level LoggingFor saving container logs to a central log store.Flexibility in ArchitectureKubernetes allows for flexibility in how these components are deployed and managed.
The architecture can be adapted to various needs, from small development environments
to large-scale production deployments.For more detailed information about each component and various ways to configure your
cluster architecture, see the Cluster Architecture page.FeedbackWas this page helpful?Yes
NoThanks for the feedback. If you have a specific, answerable question about how to use Kubernetes, ask it on Stack Overflow.
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report a problem
or
suggest an improvement.Last modified May 31, 2025 at 8:36 AM PST: Fix missing periods at the end of sentences (f0b3dc1f07)Items on this page refer to third party products or projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for those third-party products or projects. See the CNCF website guidelines for more details.You should read the content guide before proposing a change that adds an extra third-party link.